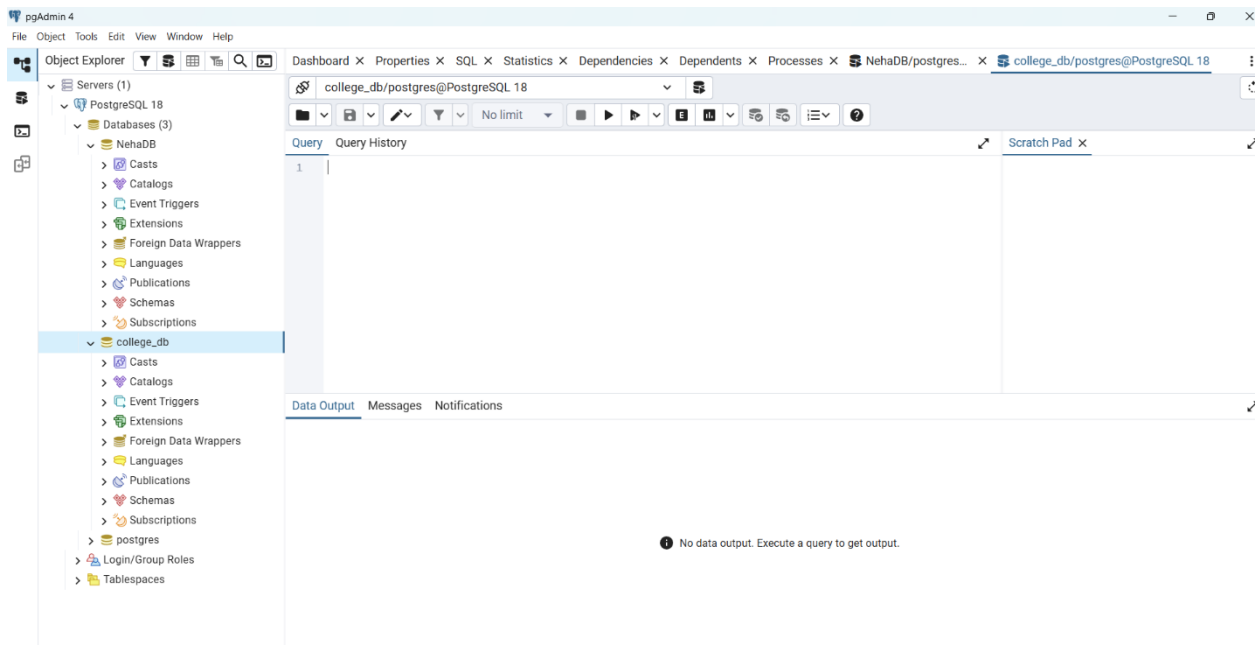


Module 3 :Database Integration HandsOn

HandsOn-1

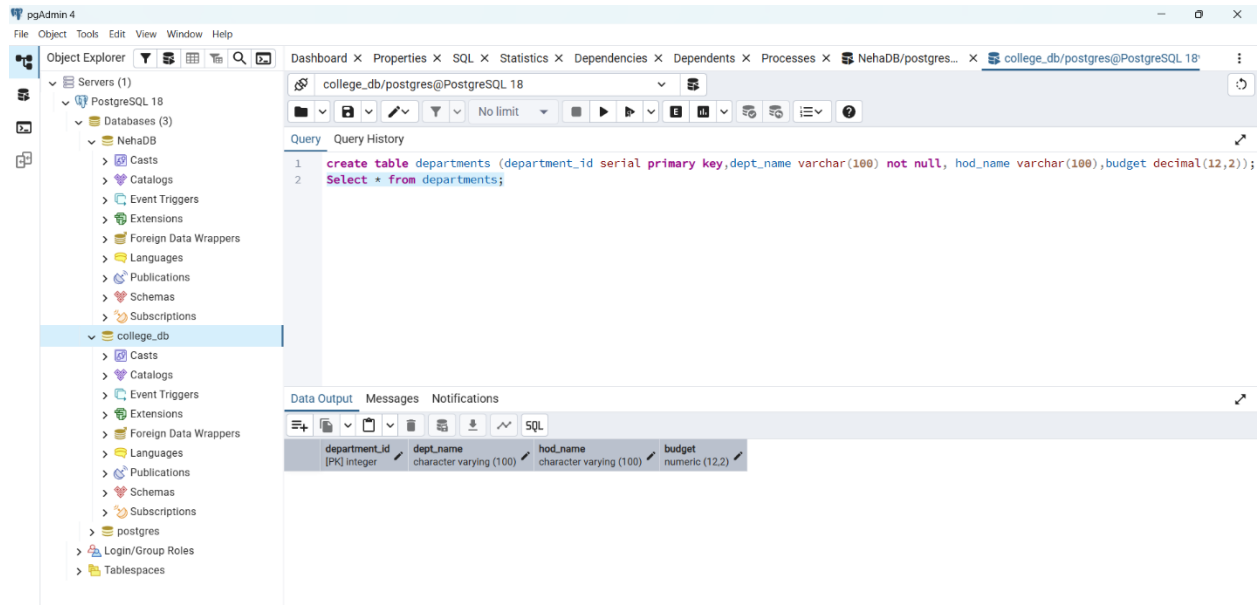
TASK 1: Create the Database and Tables Goal: Practise DDL by N creating the full college_db schema with constraints

1. Create a new database named college_db.

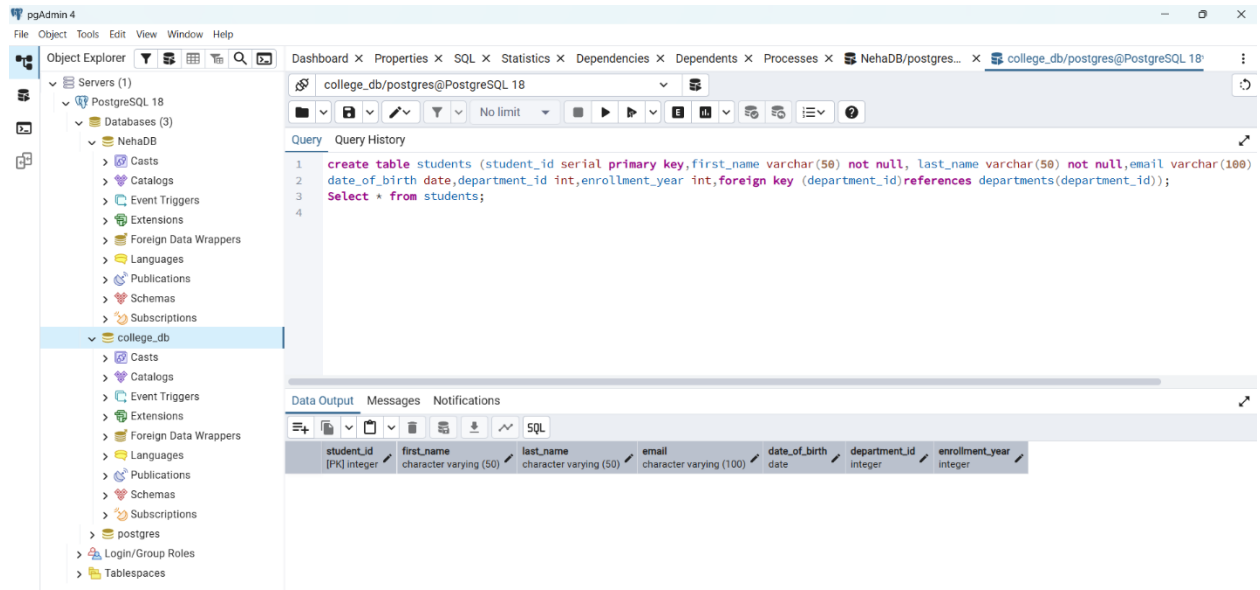


2. Write CREATE TABLE statements for all five tables listed in the schema section: departments, students, courses, enrollments, professors.

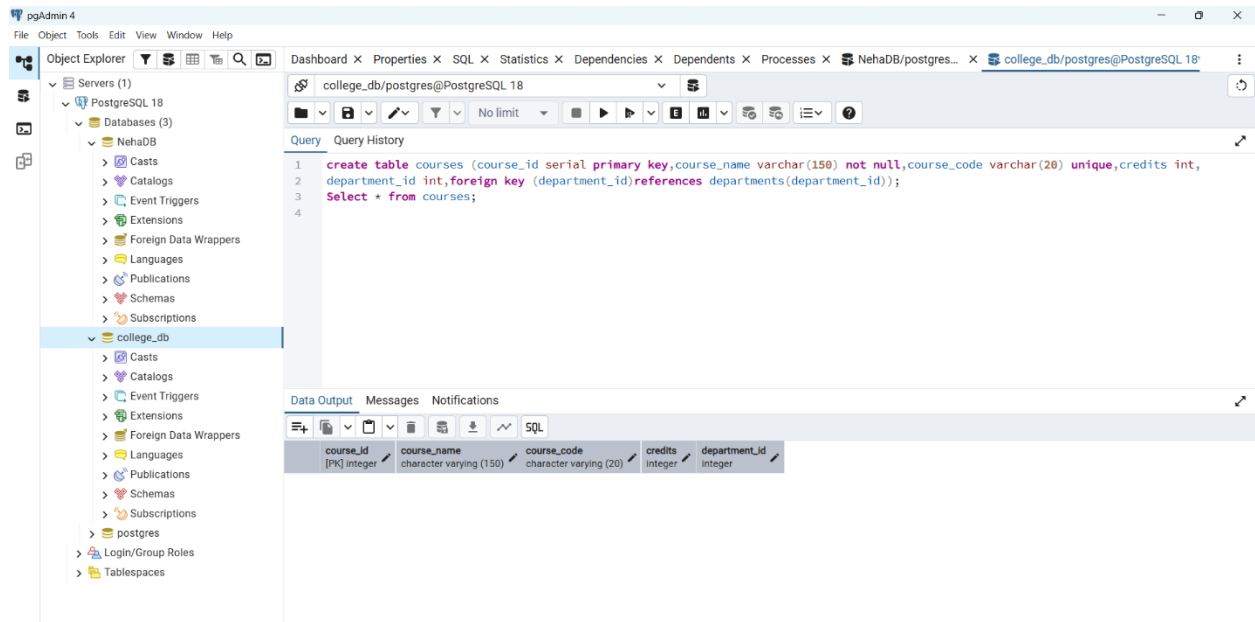
departments:



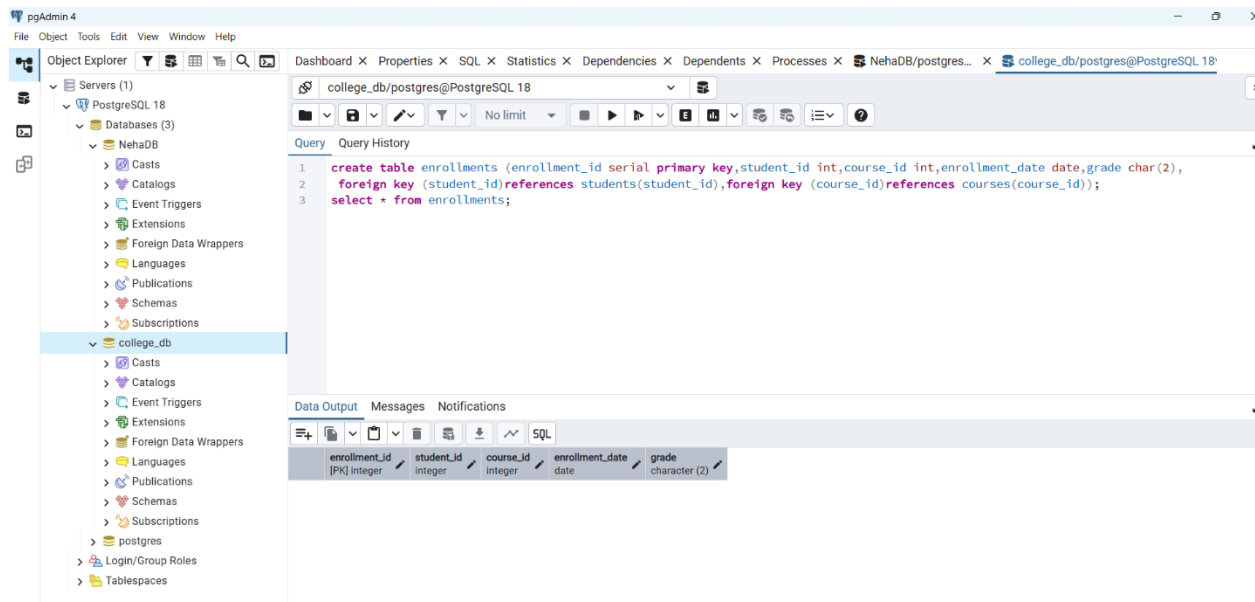
students:



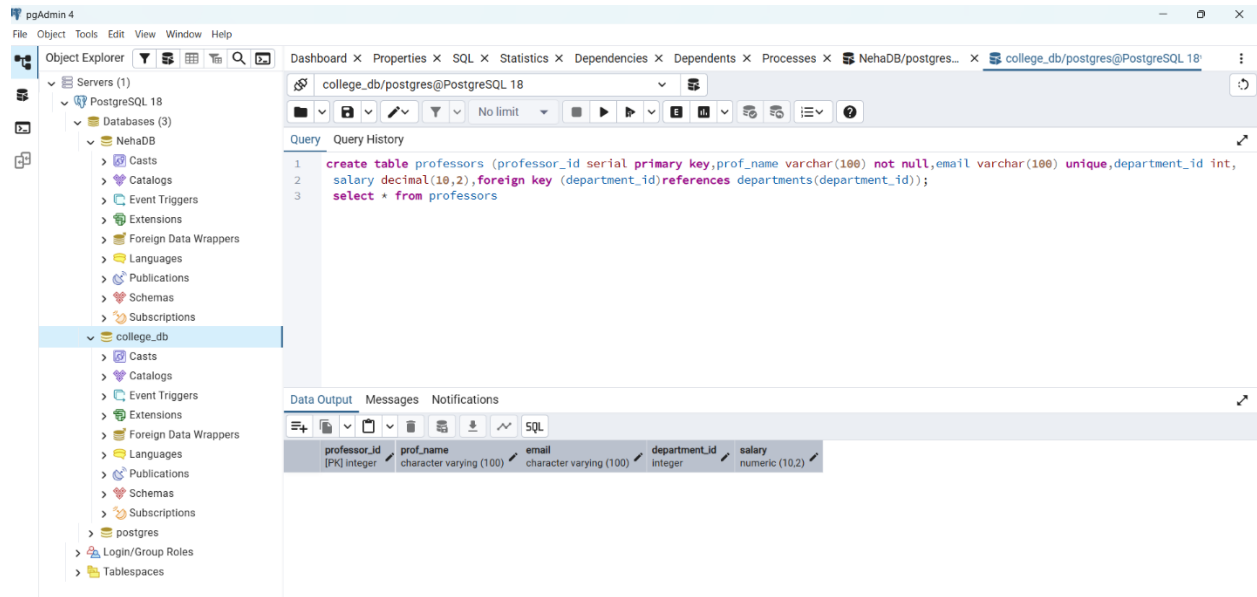
courses:



enrollments:

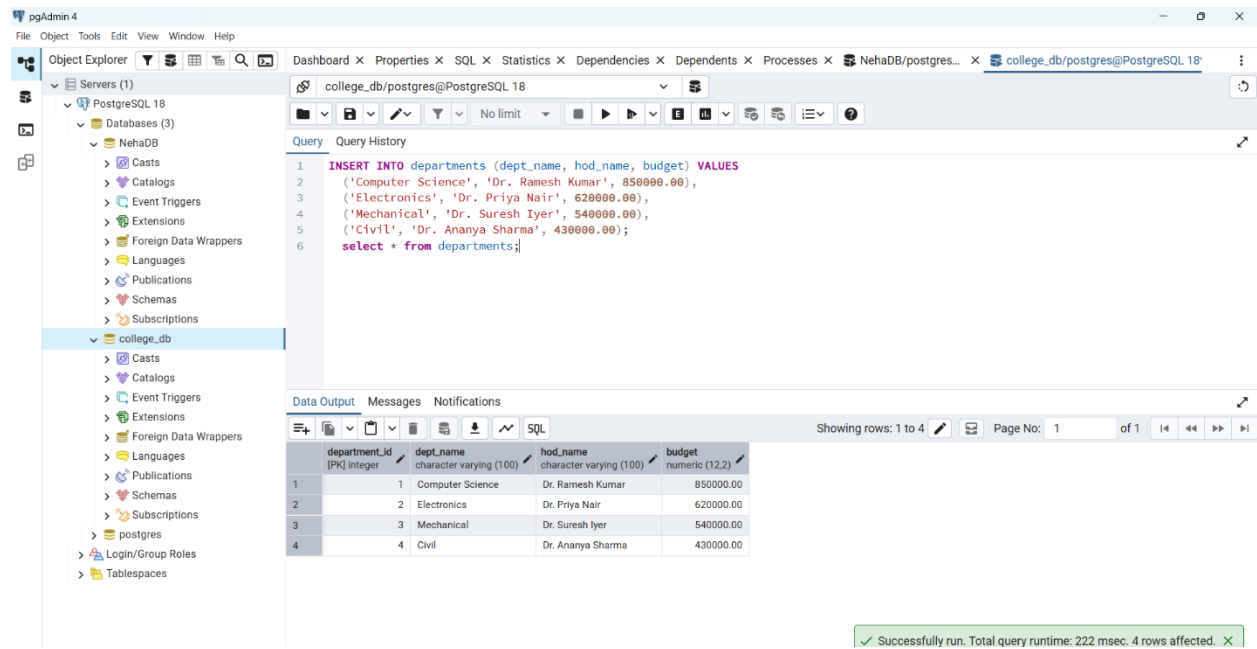


professors:



15.Insert table:

departments:



students:

pgAdmin 4

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Dashboard Properties SQL Statistics Dependencies Dependents Processes NehaDB/postgres college_db/postgres@PostgreSQL 18

college_db/postgres@PostgreSQL 18

Query Query History

```
1 INSERT INTO students (first_name, last_name, email, date_of_birth, department_id,  
2 enrollment_year) VALUES  
3 ('Arjun', 'Mehta', 'arjun.mehta@college.edu', '2003-04-12', 1, 2022),  
4 ('Priya', 'Suresh', 'priya.suresh@college.edu', '2003-07-25', 1, 2022),  
5 ('Rohan', 'Verma', 'rohan.verma@college.edu', '2002-11-08', 2, 2021),  
6 ('Sneha', 'Patel', 'sneha.patel@college.edu', '2004-01-30', 3, 2023),  
7 ('Vikram', 'Das', 'vikram.das@college.edu', '2003-09-14', 1, 2022),  
8 ('Kavya', 'Menon', 'kavya.menon@college.edu', '2002-05-17', 2, 2021),  
9 ('Aditya', 'Singh', 'aditya.singh@college.edu', '2004-03-22', 4, 2023),  
10 ('Deepika', 'Rao', 'deepika.rao@college.edu', '2003-08-09', 1, 2022);  
11 select * from students;
```

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

student_id	first_name	last_name	email	date_of_birth	department_id	enrollment_year
1	Arjun	Mehta	arjun.mehta@college.e...	2003-04-12	1	2022
2	Priya	Suresh	priya.suresh@college.e...	2003-07-25	1	2022
3	Rohan	Verma	rohan.verma@college.e...	2002-11-08	2	2021
4	Sneha	Patel	sneha.patel@college.e...	2004-01-30	3	2023
5	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022
6	Kavya	Menon	kavya.menon@college...	2002-05-17	2	2021
7	Aditya	Singh	aditya.singh@college.e...	2004-03-22	4	2023
8	Deepika	Rao	deepika.rao@college.e...	2003-08-09	1	2022

Courses:

pgAdmin 4

File Object Tools Edit View Window Help

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Dashboard Properties SQL Statistics Dependencies Dependents Processes NehaDB/postgres college_db/postgres@PostgreSQL 18

college_db/postgres@PostgreSQL 18

Query Query History

```
1 INSERT INTO courses (course_name, course_code, credits, department_id) VALUES  
2 ('Data Structures & Algorithms', 'CS101', 4, 1),  
3 ('Database Management Systems', 'CS102', 3, 1),  
4 ('Object Oriented Programming', 'CS103', 4, 1),  
5 ('Circuit Theory', 'EC101', 3, 2),  
6 ('Thermodynamics', 'ME101', 3, 3);  
7 select * from courses;
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

course_id	course_name	course_code	credits	department_id
1	Data Structures & Algorith...	CS101	4	1
2	Database Management Sy...	CS102	3	1
3	Object Oriented Programm...	CS103	4	1
4	Circuit Theory	EC101	3	2
5	Thermodynamics	ME101	3	3

Enrollments:

pgAdmin 4

Object Explorer: Servers (1) > PostgreSQL 18 > Databases (3) > NehaDB > college_db

Query: college_db/postgres@PostgreSQL 18

```

1 INSERT INTO enrollments (student_id, course_id, enrollment_date, grade) VALUES
2 (1, 1, '2022-07-01', 'A'), (1, 2, '2022-07-01', 'B'),
3 (2, 1, '2022-07-01', 'B'), (2, 3, '2022-07-01', 'A'),
4 (3, 4, '2021-07-01', 'A'), (4, 5, '2023-07-01', NULL),
5 (5, 1, '2022-07-01', 'C'), (5, 2, '2022-07-01', 'A'),
6 (6, 4, '2021-07-01', 'B'), (7, 5, '2023-07-01', NULL),
7 (8, 1, '2022-07-01', 'A'), (8, 3, '2022-07-01', 'B')
8 select * from enrollments;

```

Data Output: Showing rows: 1 to 12 | Page No: 1 of 1

enrollment_id [PK] integer	student_id integer	course_id integer	enrollment_date date	grade character (2)
1	1	1	2022-07-01	A
2	2	1	2022-07-01	B
3	3	2	2022-07-01	B
4	4	2	2022-07-01	A
5	5	3	2021-07-01	A
6	6	4	2023-07-01	[null]
7	7	5	2022-07-01	C
8	8	5	2022-07-01	A
9	9	6	2021-07-01	R

Total rows: 12 | Query complete 00:00:00.182 | CRLF | Ln 8, Col 1

professors:

pgAdmin 4

Object Explorer: Servers (1) > PostgreSQL 18 > Databases (3) > NehaDB > college_db

Query: college_db/postgres@PostgreSQL 18

```

1 INSERT INTO professors (prof_name, email, department_id, salary) VALUES
2 ('Dr. Anand Krishnan', 'anand.k@college.edu', 1, 95000.00),
3 ('Dr. Meena Pillai', 'meena.p@college.edu', 1, 88000.00),
4 ('Dr. Sunil Rajan', 'sunil.r@college.edu', 2, 82000.00),
5 ('Dr. Latha Gopal', 'latha.g@college.edu', 3, 79000.00),
6 ('Dr. Kartik Bose', 'kartik.b@college.edu', 4, 76000.00);
7 select * from professors;

```

Data Output: Showing rows: 1 to 5 | Page No: 1 of 1

professor_id [PK] integer	prof_name character varying (100)	email character varying (100)	department_id integer	salary numeric (10,2)
1	Dr. Anand Krishnan	anand.k@college.edu	1	95000.00
2	Dr. Meena Pillai	meena.p@college.edu	1	88000.00
3	Dr. Sunil Rajan	sunil.r@college.edu	2	82000.00
4	Dr. Latha Gopal	latha.g@college.edu	3	79000.00
5	Dr. Kartik Bose	kartik.b@college.edu	4	76000.00

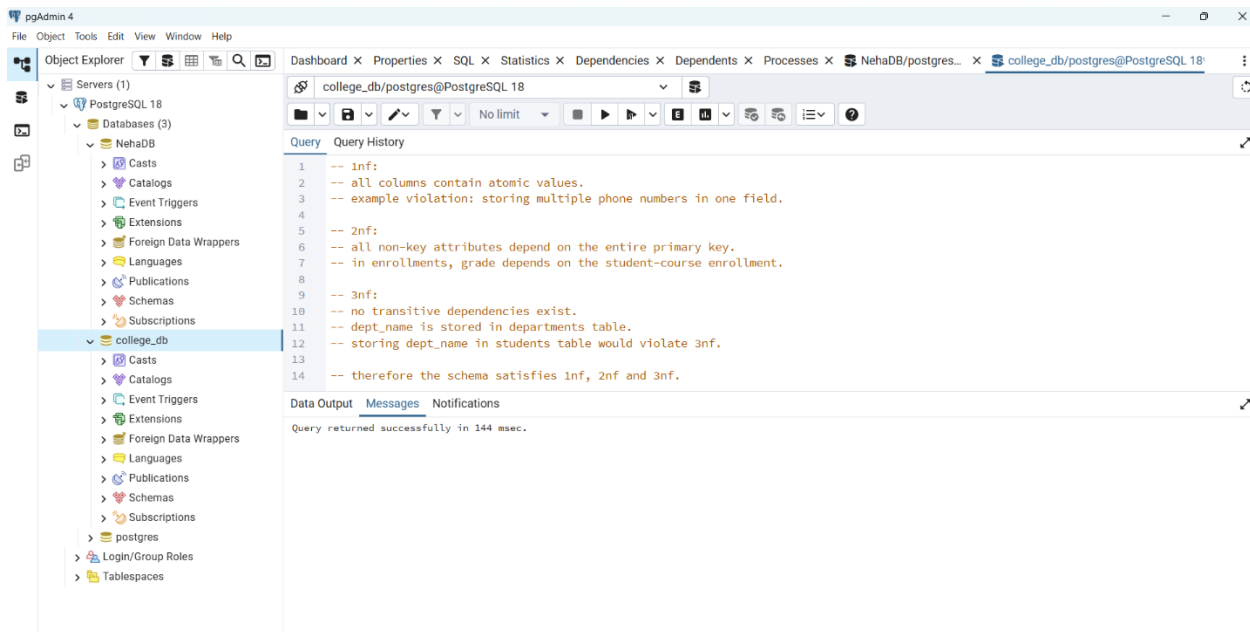
TASK 2:

6. For 1NF: Confirm every column holds atomic values. Identify any hypothetical violation if, for example, we had stored multiple phone numbers in one field.

7. For 2NF: Confirm every non-key column is fully dependent on the entire primary key (especially check the enrollments table, which has a composite candidate key of student_id + course_id).

8. For 3NF: Confirm there are no transitive dependencies. (Hint: would storing dept_name in the students table violate 3NF?)

9. Write a short comment using SQL -- comments inside your .sql file documenting your 3NF analysis for the enrollments table



TASK 3:

10. Add a column phone_number VARCHAR(15) to the students table

pgAdmin 4

Object Explorer: Servers (1) > PostgreSQL 18 > Databases (3) > NehaDB > college_db

Query:

```

1 alter table students
2 add column phone_number varchar(15);
3 select * from students;

```

Data Output:

student_id [PK] integer	first_name character varying (50)	last_name character varying (50)	email character varying (100)	date_of_birth date	department_id integer	enrollment_year integer	phone_number character varying (15)
1	Arjun	Mehra	arjun.mehra@college.e...	2003-04-12	1	2022	[null]
2	Priya	Suresh	priya.suresh@college.e...	2003-07-25	1	2022	[null]
3	Rohan	Verma	rohan.verma@college.e...	2002-11-08	2	2021	[null]
4	Sneha	Patel	sneha.patel@college.e...	2004-01-30	3	2023	[null]
5	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022	[null]
6	Kavya	Menon	kavya.menon@college...	2002-05-17	2	2021	[null]
7	Aditya	Singh	aditya.singh@college.e...	2004-03-22	4	2023	[null]
8	Deepika	Rao	deepika.rao@college.e...	2003-08-09	1	2022	[null]

11. Add a column max_seats INT DEFAULT 60 to the courses table.

pgAdmin 4

Object Explorer: Servers (1) > PostgreSQL 18 > Databases (3) > NehaDB > college_db

Query:

```

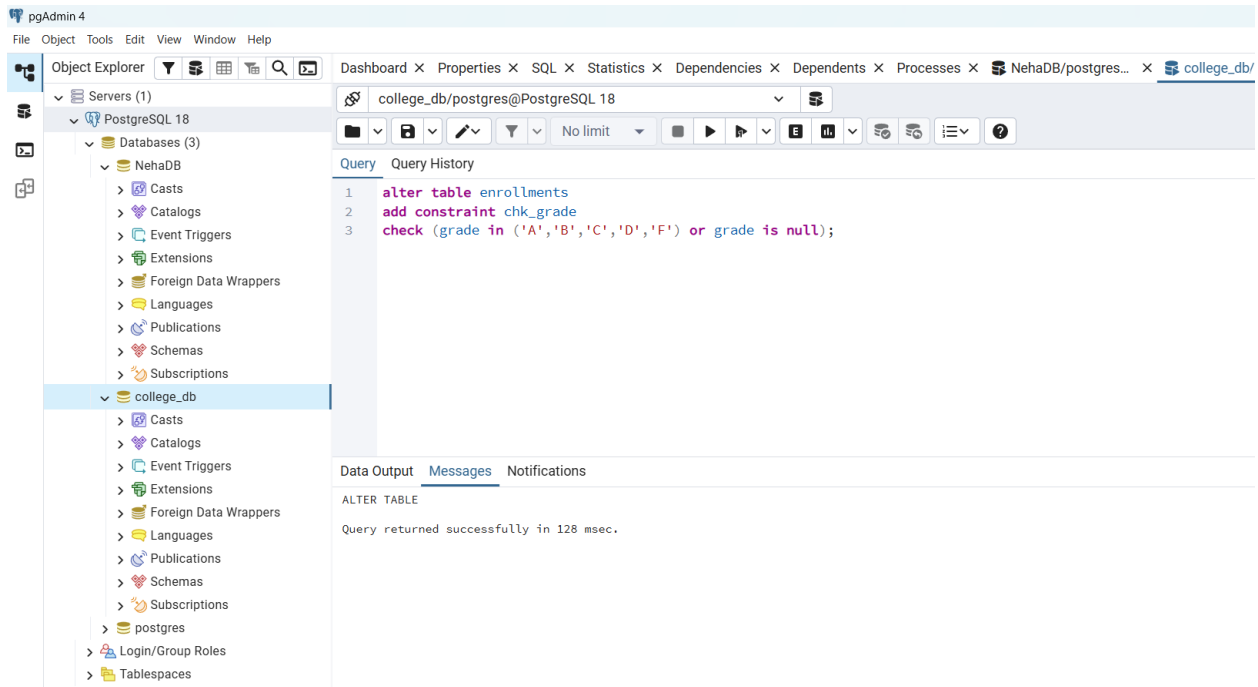
1 alter table courses
2 add column max_seats int default 60;
3 select * from courses;

```

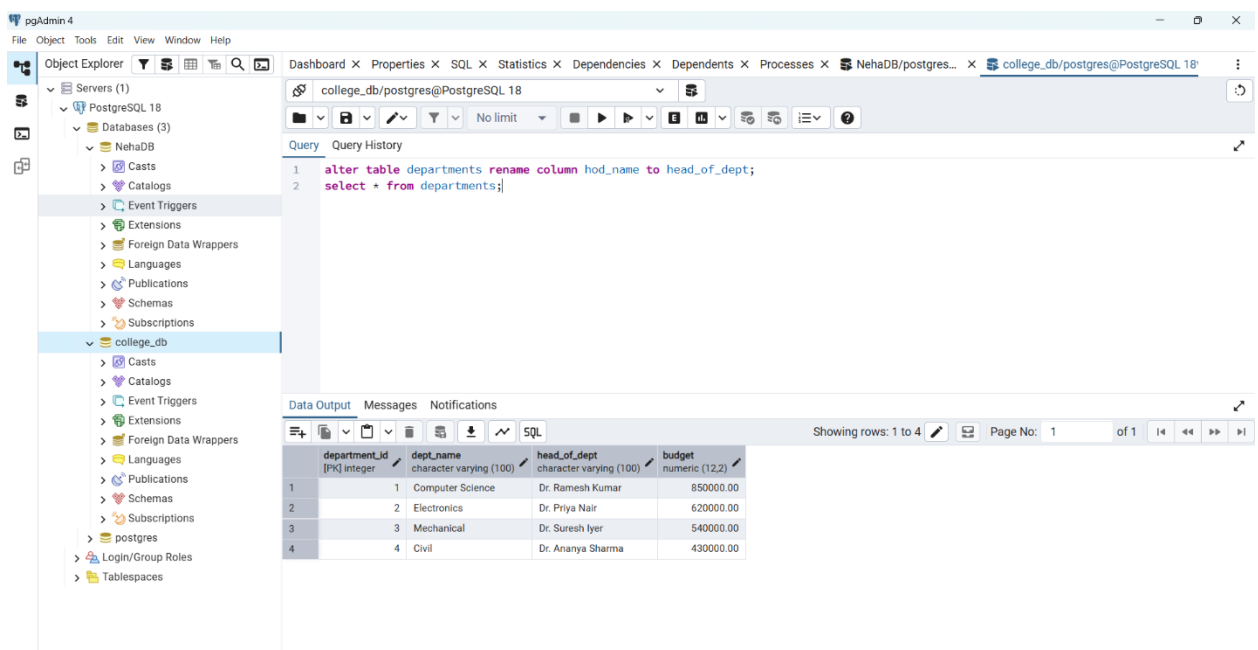
Data Output:

course_id [PK] integer	course_name character varying (150)	course_code character varying (20)	credits integer	department_id integer	max_seats integer
1	Data Structures & Algorith...	CS101	4	1	60
2	Database Management Sy...	CS102	3	1	60
3	Object Oriented Programm...	CS103	4	1	60
4	Circuit Theory	EC101	3	2	60
5	Thermodynamics	ME101	3	3	60

12. Add a CHECK constraint to enrollments ensuring grade is one of ('A','B','C','D','F') or NULL



13. Rename the hod_name column in departments to head_of_dept (PostgreSQL: ALTER COLUMN ... RENAME TO; MySQL: ALTER TABLE ... CHANGE ...).



14. Drop the phone_number column you added in step 1 (simulate a schema rollback)

pgAdmin 4

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Dashboard x Properties x SQL x Statistics x Dependencies x Dependents x Processes x NehaDB/postgres... x college_db/postgres@PostgreSQL 18

college_db/postgres@PostgreSQL 18

Query Query History

```

1 alter table students drop column phone_number;
2 select * from students;

```

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

	student_id [PK] integer	first_name character varying (50)	last_name character varying (50)	email character varying (100)	date_of_birth date	department_id integer	enrollment_year integer
1	1	Arjun	Mehta	arjun.mehta@college.e...	2003-04-12	1	2022
2	2	Priya	Suresh	priya.suresh@college.e...	2003-07-25	1	2022
3	3	Rohan	Verma	rohan.verma@college.e...	2002-11-08	2	2021
4	4	Sneha	Patel	sneha.patel@college.e...	2004-01-30	3	2023
5	5	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022
6	6	Kavya	Menon	kavya.menon@college...	2002-05-17	2	2021
7	7	Aditya	Singh	aditya.singh@college.e...	2004-03-22	4	2023
8	8	Deepika	Rao	deepika.rao@college.e...	2003-08-09		

✓ Successfully run. Total query runtime: 191 msec. 8 row

Total rows: 8 Query complete 00:00:00.191

HandsOn-2

TASK 1:

16. Insert two additional students of your own choosing into the students table.

pgAdmin 4

Object Explorer: Servers (1) > PostgreSQL 18 > Databases (3) > NehaDB > college_db

Query:

```
1 insert into students
2 (first_name,last_name,email,date_of_birth,department_id,enrollment_year)
3 values
4 ('rahul','sharma','rahul.sharma@college.edu','2003-05-10',1,2022),
5 ('anita','reddy','anita.reddy@college.edu','2004-02-15',2,2023);
6 select * from students;
```

Data Output:

student_id	first_name	last_name	email	date_of_birth	department_id	enrollment_year
4	Pritya	Suresh	pritya.suresh@college.edu	2003-07-25	1	2022
3	Rohan	Verma	rohan.verma@college.edu	2002-11-08	2	2021
4	Sneha	Patel	sneha.patel@college.edu	2004-01-30	3	2023
5	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022
6	Kavya	Menon	kavya.menon@college.edu	2002-05-17	2	2021
7	Aditya	Singh	aditya.singh@college.edu	2004-03-22	4	2023
8	Deepika	Rao	deepika.rao@college.edu	2003-08-09	1	2022
9	rahul	sharma	rahul.sharma@college.edu	2003-05-10	1	2022
10	anita	reddy	anita.reddy@college.edu	2004-02-15	2	2023

17. Update the grade of student_id = 5 for course_id = 1 from 'C' to 'B'

pgAdmin 4

Object Explorer: Servers (1) > PostgreSQL 18 > Databases (3) > NehaDB > college_db

Query:

```
1 select count(*) from students;
```

Data Output:

count
10

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer Servers (1) PostgreSQL 18 Databases (3) NehaDB college_db

college_db/postgres@PostgreSQL 18

Query

```
1 update enrollments set grade='B' where student_id=5 and course_id=1;
2 select * from enrollments where student_id=5 and course_id=1;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

enrollment_id	student_id	course_id	enrollment_date	grade
1	7	5	2022-07-01	B

18. Delete enrollments where grade IS NULL (students who never received a grade)

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer Servers (1) PostgreSQL 18 Databases (3) NehaDB college_db

college_db/postgres@PostgreSQL 18

Query

```
1 select * from enrollments where grade is null;
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

enrollment_id	student_id	course_id	enrollment_date	grade
1	6	4	2023-07-01	[null]
2	10	7	2023-07-01	[null]

pgAdmin 4

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college_db/postgres@PostgreSQL 18

Query

```
1 select * from enrollments where grade is null;
2 delete from enrollments where grade is null;
3 select * from enrollments;
```

Data Output

enrollmentId [PK] integer	studentId integer	courseId integer	enrollment_date date	grade character (2)
1	1	1	2022-07-01	A
2	2	1	2022-07-01	B
3	3	2	2022-07-01	B
4	4	2	2022-07-01	A
5	5	3	2021-07-01	A
6	8	5	2022-07-01	A
7	9	6	2021-07-01	B
8	11	8	2022-07-01	A
9	12	8	2022-07-01	B

Total rows: 10 Query completed: 00:00:00.100

19. Verify row counts using SELECT COUNT(*) after each operation

pgAdmin 4

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college_db/postgres@PostgreSQL 18

Query

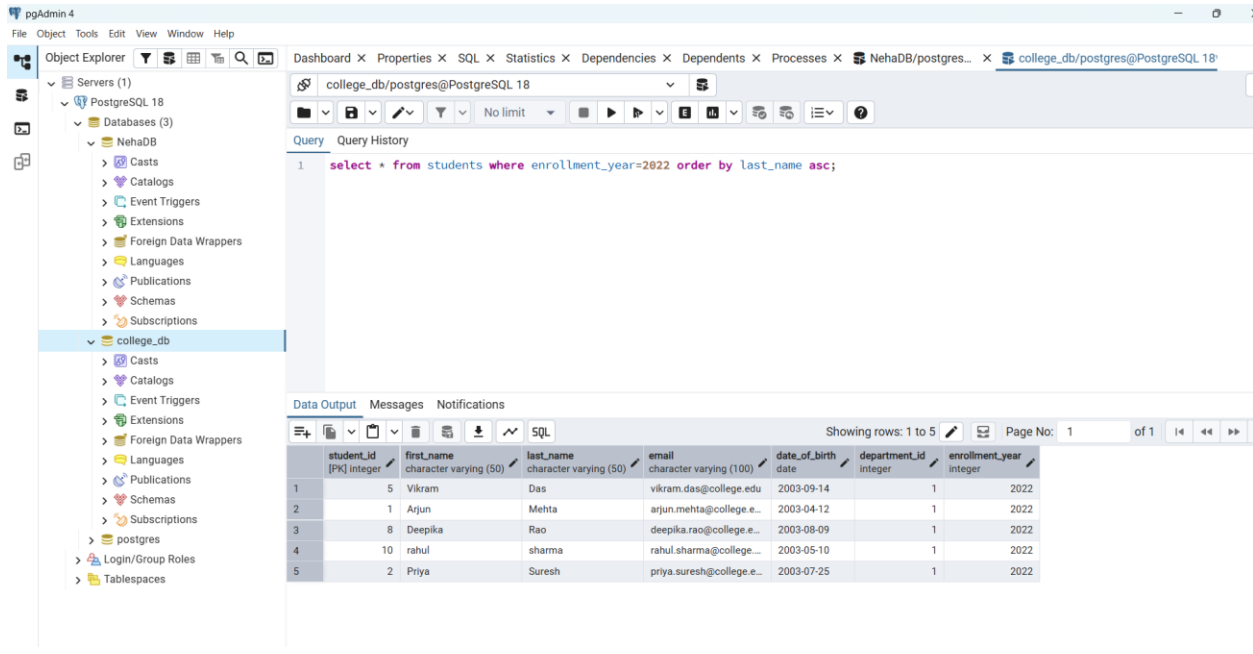
```
1 select count(*) from students;
2 select count(*) from enrollments;
```

Data Output

count bigint
10

TASK 2: Single-Table Queries and Filtering

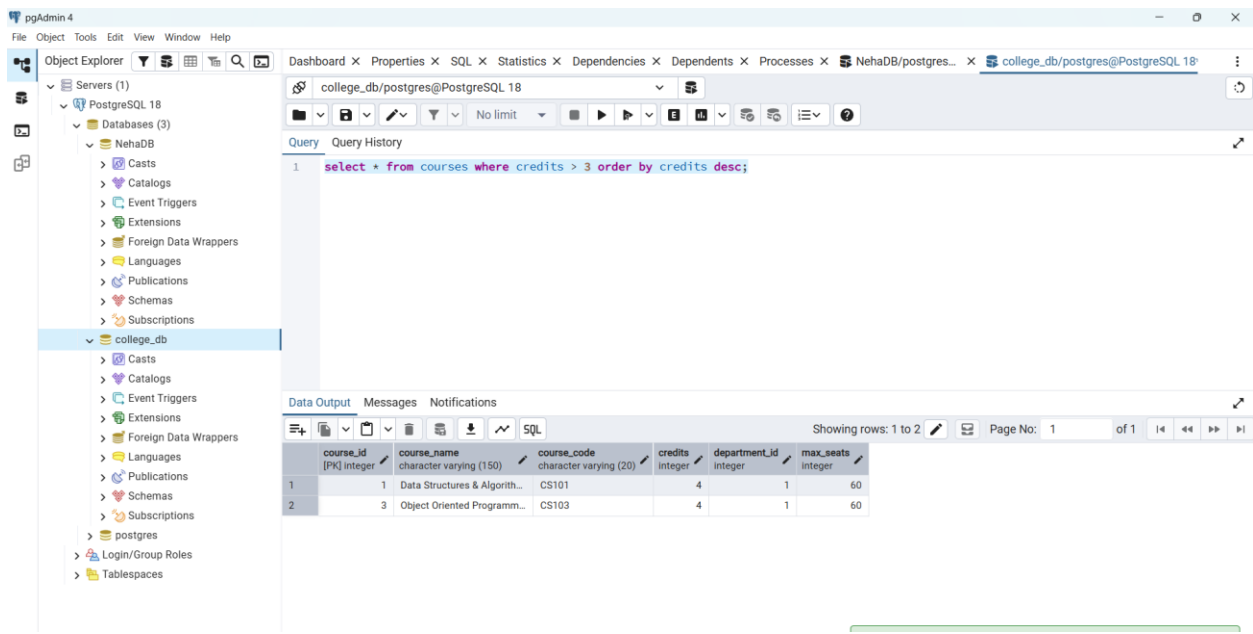
20. Retrieve all students enrolled in 2022, ordered by last_name alphabetically



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected. The main pane shows a SQL query: `select * from students where enrollment_year=2022 order by last_name asc;`. The 'Data Output' tab displays the results of the query, showing 5 rows of student data.

student_id [PK] integer	first_name character varying (50)	last_name character varying (50)	email character varying (100)	date_of_birth date	department_id integer	enrollment_year integer
1	5 Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022
2	1 Arjun	Mehta	arjun.mehta@college.e...	2003-04-12	1	2022
3	8 Deepika	Rao	deepika.rao@college.e...	2003-08-09	1	2022
4	10 rahul	sharma	rahul.sharma@college...	2003-05-10	1	2022
5	2 Priya	Suresh	priya.suresh@college.e...	2003-07-25	1	2022

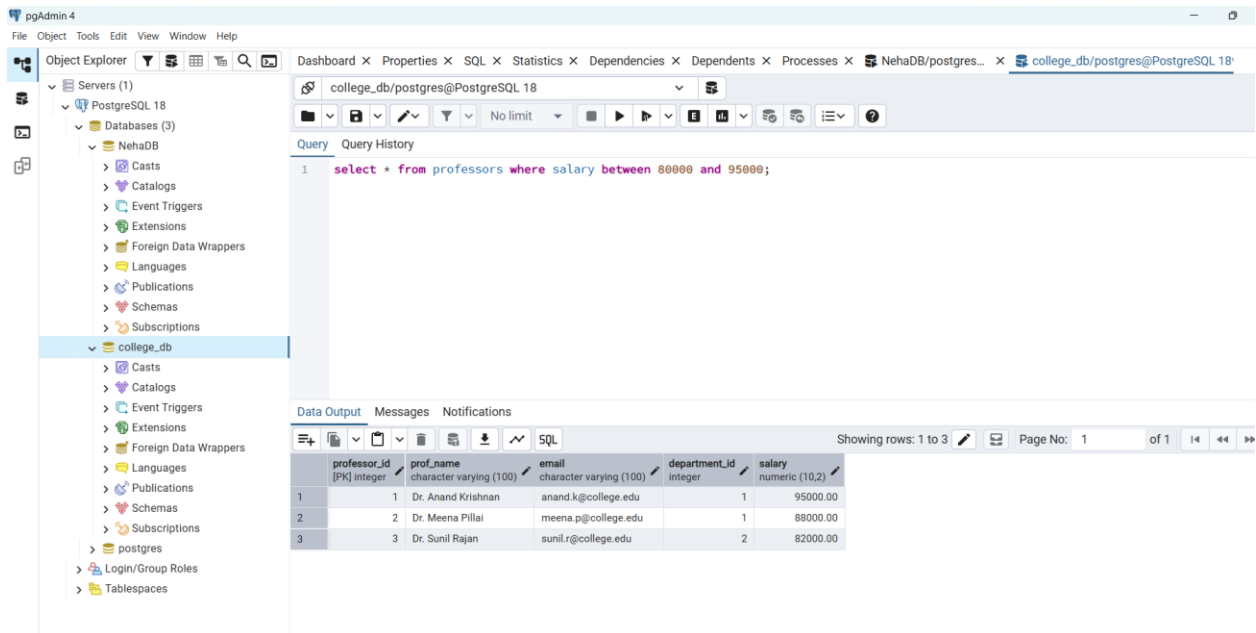
21. Find all courses with more than 3 credits, sorted by credits descending



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected. The main pane shows a SQL query: `select * from courses where credits > 3 order by credits desc;`. The 'Data Output' tab displays the results of the query, showing 2 rows of course data.

course_id [PK] integer	course_name character varying (150)	course_code character varying (20)	credits integer	department_id integer	max_seats integer
1	Data Structures & Algorith...	CS101	4	1	60
2	Object Oriented Programm...	CS103	4	1	60

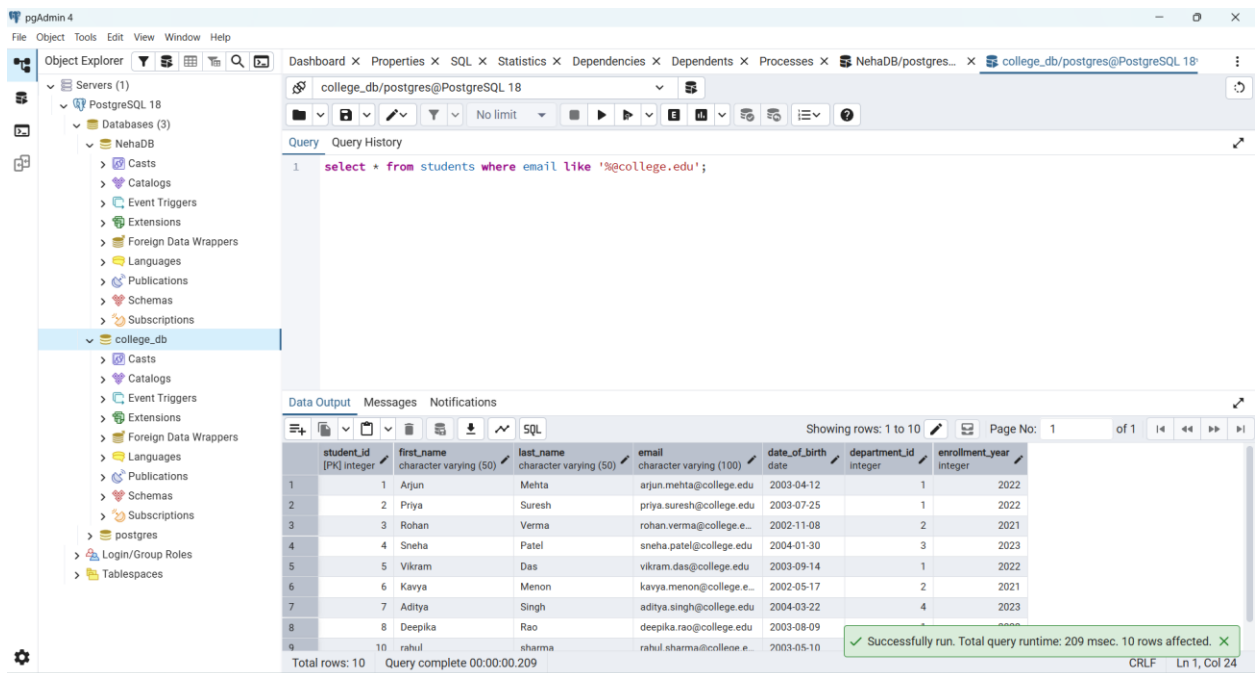
22. List all professors whose salary is between 80,000 and 95,000.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected under 'PostgreSQL 18'. The main pane shows a SQL query: `select * from professors where salary between 80000 and 95000;`. The 'Data Output' tab is active, displaying a table with 3 rows of professor data.

professor_id [PK] integer	prof_name character varying (100)	email character varying (100)	department_id integer	salary numeric (10,2)
1	Dr. Anand Krishnan	anand.k@college.edu	1	95000.00
2	Dr. Meena Pillai	meena.p@college.edu	1	88000.00
3	Dr. Sunil Rajan	sunil.r@college.edu	2	82000.00

23. Find all students whose email ends with '@college.edu' using the LIKE operator.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected under 'PostgreSQL 18'. The main pane shows a SQL query: `select * from students where email like '%@college.edu';`. The 'Data Output' tab is active, displaying a table with 10 rows of student data. A status message at the bottom indicates the query was successfully run, affecting 10 rows.

student_id [PK] integer	first_name character varying (50)	last_name character varying (50)	email character varying (100)	date_of_birth date	department_id integer	enrollment_year integer
1	Arjun	Mehta	arjun.mehta@college.edu	2003-04-12	1	2022
2	Priya	Suresh	priya.suresh@college.edu	2003-07-25	1	2022
3	Rohan	Verma	rohan.verma@college.e...	2002-11-08	2	2021
4	Sneha	Patel	sneha.patel@college.edu	2004-01-30	3	2023
5	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022
6	Kavya	Menon	kavya.menon@college.e...	2002-05-17	2	2021
7	Aditya	Singh	aditya.singh@college.edu	2004-03-22	4	2023
8	Deepika	Rao	deepika.rao@college.edu	2003-08-09		
9						
10	rahul	sharma	rahul.sharma@college.e...	2003-05-10		

Successfully run. Total query runtime: 209 msec. 10 rows affected.

24. Count the total number of students per enrollment_year.

pgAdmin 4

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Query Query History

```
1 select enrollment_year,count(*) as total_students from students group by enrollment_year;
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	enrollment_year integer	total_students bigint
1	2022	5
2	2023	3
3	2021	2

Task 3: Multi-Table Joins

25. List each student's full name (first_name + ' ' + last_name) alongside the name of their department. (JOIN students and departments.)

pgAdmin 4

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No limit

Query Query History

```
1 select s.first_name || ' ' || s.last_name as student_name, d.dept_name from students s join departments d
2 on s.department_id=d.department_id;
```

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1 of 1

	student_name text	dept_name character varying (100)
1	Arjun Mehta	Computer Science
2	Priya Suresh	Computer Science
3	Rohan Verma	Electronics
4	Sneha Patel	Mechanical
5	Vikram Das	Computer Science
6	Kavya Menon	Electronics
7	Aditya Singh	Civil
8	Deepika Rao	Computer Science
9	rahul sharma	Computer Science

Total rows: 10 Query complete 00:00:00.210

Successfully run. Total query runtime: 210 msec. 10 rows affected

CRLF Ln 2, 0

26. Show each enrollment along with the student's name and the course name. (3-table JOIN: enrollments, students, courses.)

The screenshot shows the pgAdmin 4 interface with a query window open. The query is a 3-table JOIN between enrollments, students, and courses. The results are displayed in a table with 10 rows.

```
1 select
2 e.enrollment_id,
3 s.first_name || ' ' || s.last_name as student_name,
4 c.course_name,
5 e.grade
6 from enrollments e
7 join students s
8 on e.student_id=s.student_id
9 join courses c
10 on e.course_id=c.course_id;
```

enrollment_id	student_name	course_name	grade
1	Arjun Mehta	Data Structures & Algorith...	A
2	Arjun Mehta	Database Management Sy...	B
3	Priya Suresh	Data Structures & Algorith...	B
4	Priya Suresh	Object Oriented Program...	A
5	Rohan Verma	Circuit Theory	A
6	Vikram Das	Database Management Sy...	A
7	Kavya Menon	Circuit Theory	B
8	Deepika Rao	Data Structures & Algorith...	A
9	Deenika Rao	Object Oriented Program...	B
10			

Successfully run. Total query runtime: 155 msec. 10 rows affected.

27. Find all students who are NOT enrolled in any course using a LEFT JOIN and WHERE ... IS NULL pattern.

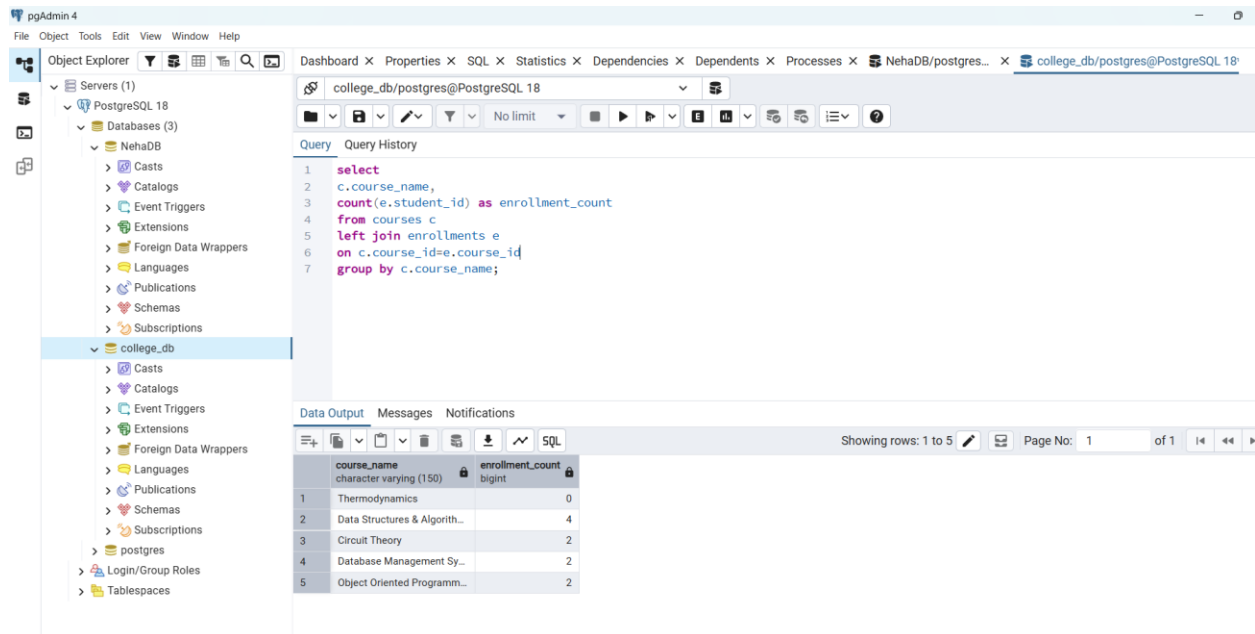
The screenshot shows the pgAdmin 4 interface with a query window open. The query is a LEFT JOIN between students and enrollments, filtering for students who are not enrolled in any course. The results are displayed in a table with 4 rows.

```
1 select s.student_id, s.first_name, s.last_name from students s left join enrollments e on s.student_id=e.student_id
2 where e.student_id is null;
```

student_id	first_name	last_name
11	anita	reddy
10	rahul	sharma
4	Sneha	Patel
7	Aditya	Singh

Successfully run. Total query runtime: 232 msec. 4 rows affected.

28. Display every course along with the number of students enrolled in it. Courses with zero enrolments must still appear. (LEFT JOIN courses with enrollments, GROUP BY course.)

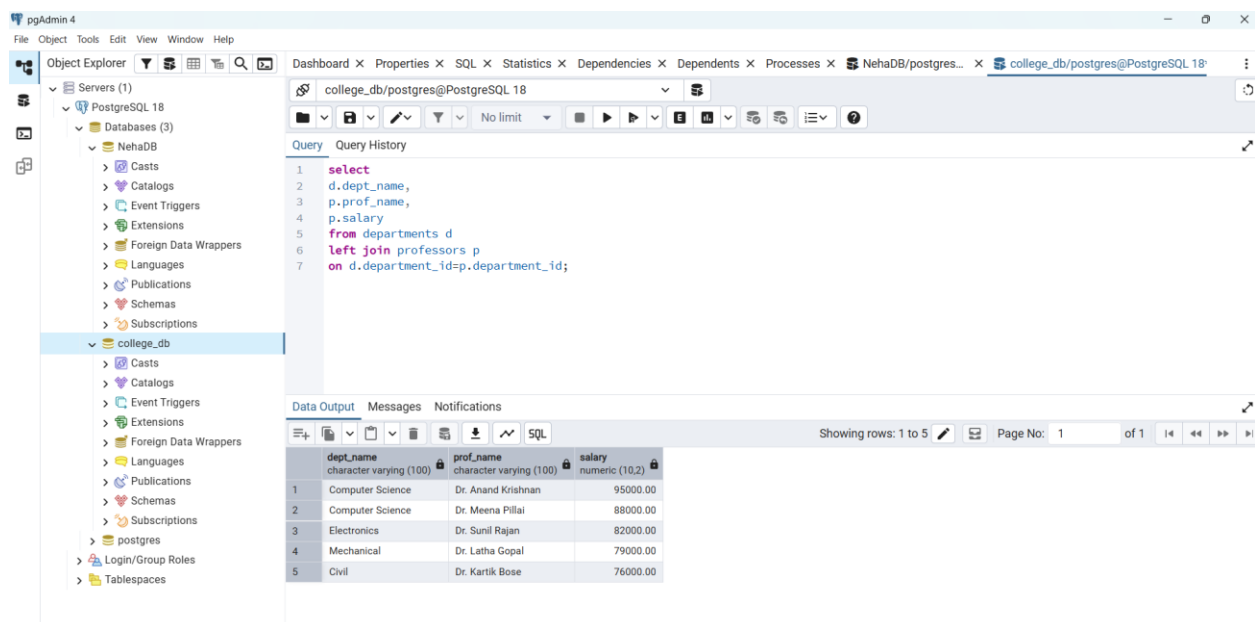


The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'college_db' database. The main pane shows a SQL query that uses a LEFT JOIN between the 'courses' and 'enrollments' tables, grouped by course name. The 'Data Output' tab at the bottom displays the results of the query.

```
1 select
2   c.course_name,
3   count(e.student_id) as enrollment_count
4 from courses c
5 left join enrollments e
6 on c.course_id=e.course_id
7 group by c.course_name;
```

course_name	enrollment_count
Thermodynamics	0
Data Structures & Algorith...	4
Circuit Theory	2
Database Management Sy...	2
Object Oriented Programm...	2

29. List each department along with its professors and their salaries. Include departments that have no professors yet.



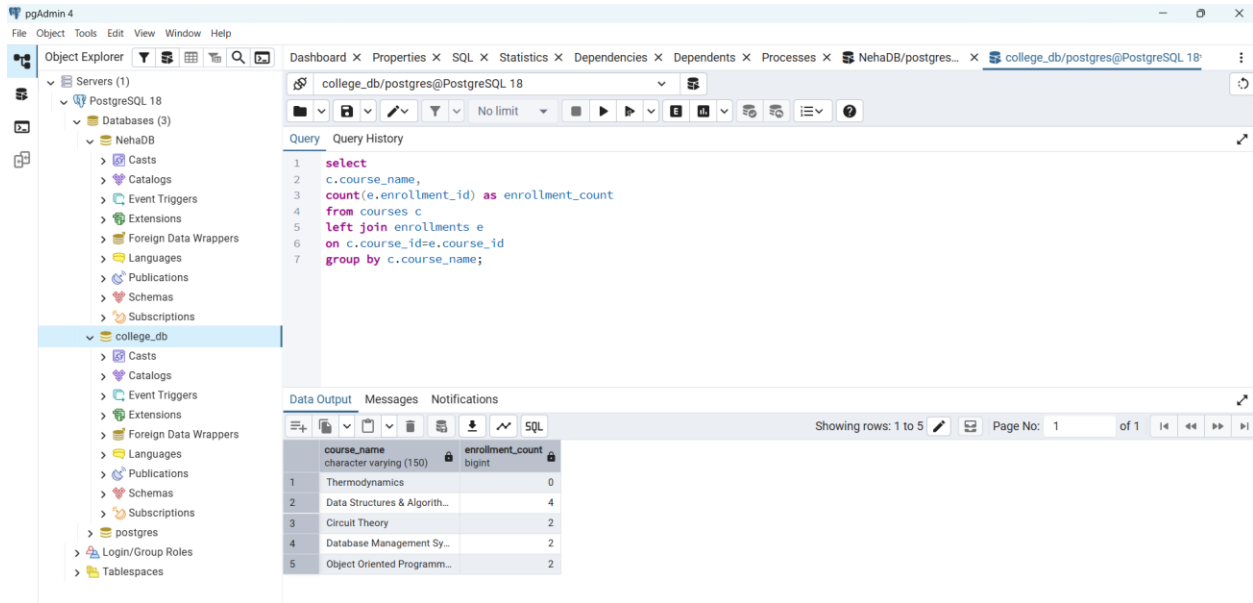
The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'college_db' database. The main pane shows a SQL query that uses a LEFT JOIN between the 'departments' and 'professors' tables. The 'Data Output' tab at the bottom displays the results of the query.

```
1 select
2   d.dept_name,
3   p.prof_name,
4   p.salary
5 from departments d
6 left join professors p
7 on d.department_id=p.department_id;
```

dept_name	prof_name	salary
Computer Science	Dr. Anand Krishnan	95000.00
Computer Science	Dr. Meena Pillai	88000.00
Electronics	Dr. Sunil Rajan	82000.00
Mechanical	Dr. Latha Gopal	79000.00
Civil	Dr. Kartik Bose	76000.00

TASK 4:

30. Calculate the total number of enrollments per course. Display course_name and enrollment_count.



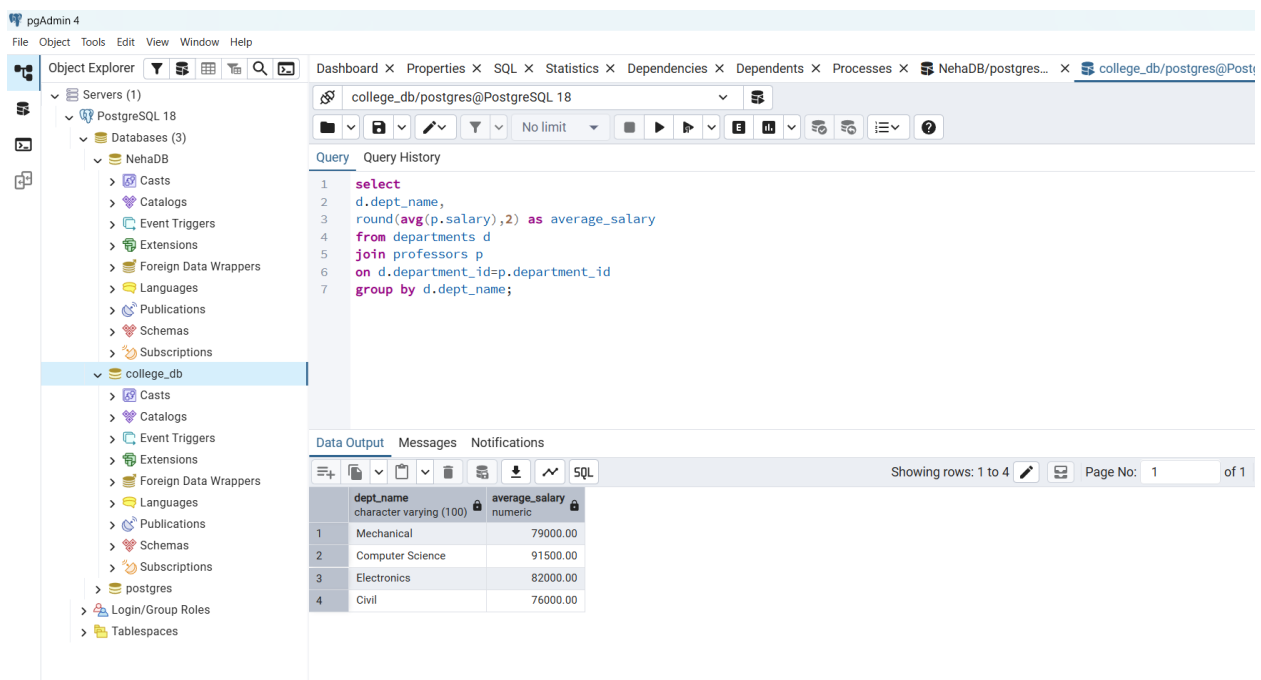
The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected. The main pane shows a SQL query in the 'Query' tab:

```
1 select
2 c.course_name,
3 count(e.enrollment_id) as enrollment_count
4 from courses c
5 left join enrollments e
6 on c.course_id=e.course_id
7 group by c.course_name;
```

The 'Data Output' tab shows the results of the query:

course_name	enrollment_count
Thermodynamics	0
Data Structures & Algorith...	4
Circuit Theory	2
Database Management Sy...	2
Object Oriented Programm...	2

31. Find the average salary of professors per department. Round to 2 decimal places.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'college_db' selected. The main pane shows a SQL query in the 'Query' tab:

```
1 select
2 d.dept_name,
3 round(avg(p.salary),2) as average_salary
4 from departments d
5 join professors p
6 on d.department_id=p.department_id
7 group by d.dept_name;
```

The 'Data Output' tab shows the results of the query:

deptName	average_salary
Mechanical	79000.00
Computer Science	91500.00
Electronics	82000.00
Civil	76000.00

32. Find all departments where the total budget exceeds 600,000.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with 'college_db' selected. The central pane shows a SQL query:

```
1 select *
2 from departments
3 where budget > 600000;
```

The bottom pane shows the query results in a table with 4 columns: department_id, dept_name, head_of_dept, and budget. The results are as follows:

department_id	dept_name	head_of_dept	budget
1	Computer Science	Dr. Ramesh Kumar	850000.00
2	Electronics	Dr. Priya Nair	620000.00

33. Show the grade distribution for course CS101: count of each grade (A, B, C, D, F).

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with 'college_db' selected. The central pane shows a SQL query:

```
1 select
2 e.grade,
3 count(*) as total
4 from enrollments e
5 join courses c
6 on e.course_id=c.course_id
7 where c.course_code='CS101'
8 group by e.grade;
```

The bottom pane shows the query results in a table with 2 columns: grade and total. The results are as follows:

grade	total
A	2
B	2

34. Using HAVING, list departments where more than 2 students are enrolled across all courses in that department.

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No limit

Query Query History

```
1 select
2 d.dept_name,
3 count(s.student_id) as total_students
4 from departments d
5 join students s
6 on d.department_id=s.department_id
7 group by d.dept_name
8 having count(s.student_id) > 2;
```

Data Output Messages Notifications

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	dept_name character varying (100)	total_students bigint
1	Computer Science	5
2	Electronics	3